IMPORTANT NOTES

SOME IMPORTANT INTEGRAL FORMULA

1. $\int dx = x + C$
2. $\int x^n dx = \frac{x^{n+1}}{n+1} + C (n \neq -1)$
3. $\int e^x dx = e^x + C$
4. $\int \frac{dx}{x} = \ln |x| + C$
5. $\int a^x dx = \frac{1}{\ln a} a^x + C$
6. $\int \sin x \, dx = -\cos x + C$
7. $\int \cos x \, dx = \sin x + C$
8. $\int \sec^2 x \, dx = \tan x + C$
9. $\int \text{cosec}^2 x \, dx = -\cot x + C$
10. $\int \sec x \tan x \, dx = \sec x + C$
11. $\int \text{cosec}\, x \cot x \, dx = -\text{cosec}\, x + C$
12. $\int \frac{dx}{\sqrt{a^2 - x^2}} = arc \sin \frac{x}{a} + C$
13. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a} arc \tan \frac{x}{a} + C$
14. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} arc \sec \frac{|x|}{a} + C$
15. $\int \frac{dx}{ax+b} = \frac{1}{a} \ln |ax+b|$
16. $\int (ax+b)^n dx = \frac{1}{a(n+1)} (ax+b)^{n+1}$
17. $\int \frac{xdx}{(ax+b)^2} = \frac{b}{a^2(ax+b)} + \frac{1}{a^2} \ln |ax+b|$
18. $\int \frac{dx}{x(ax+b)} = \frac{1}{b} \ln \frac{x}{ax+b}$
19. $\int \frac{dx}{x^2(ax+b)} = \frac{1}{bx} + \frac{a}{b^2} \ln \left| \frac{ax+b}{x} \right|$
20. $\int \sqrt{sx+b} \, dx = \frac{2}{3a} \sqrt{(ax+b)^3}$
21. $\int x\sqrt{ax+b} dx = \frac{2(3ax-2b)}{15a^2} \sqrt{(ax+b)^3}$
22. $\int \frac{dx}{\sqrt{ax+b}} = \frac{2\sqrt{ax+b}}{a}$
23. Integration by parts $\int udv = uv - \int vdu$

SOME IMPORTANT DIFFERENTIATION FORMULA

1. $\frac{d}{dx}(x) = 1$
2. $\frac{d}{dx}(ax) = a$
3. $\frac{d}{dx}(x^n) = nx^{n-1}$
4. $\frac{d}{dx}(\cos x) = -\sin x$
5. $\frac{d}{dx}(\sin x) = \cos x$
6. $\frac{d}{dx}(\tan x) = \sec^2 x$
7. $\frac{d}{dx}(\cot x) = -\text{cosec}^2 x$
8. $\frac{d}{dx}(\sec x) = \sec x \tan x$
9. $\frac{d}{dx}(\text{cosec}\, x) = -\text{cosec}\, x \cot x$
10. $\frac{d}{dx}(\ln x) = \frac{1}{x}$
11. $\frac{d}{dx}(e^x) = e^x$
12. $\frac{d}{dx}(a^x) = (\ln a) a^x$
13. $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
14. $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
15. $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$
16. Product rule $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$

SOME STANDARD INTEGRAL FORMULA FOR APPLICATION IN PHYSICS

- $\int_0^\infty e^{-ax} dx = \frac{1}{a} (\text{Re}(a) > 0)$
- $\int_0^\infty e^{-ax^2} dx = \frac{1}{2}\sqrt{\frac{\pi}{a}} (a > 0)$
- $\int_{-\infty}^\infty e^{-ax^2} dx = \sqrt{\frac{\pi}{a}} (a > 0)$
- $\int_{-\infty}^\infty e^{-ax^2+bx} dx = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}}$
- $\int_{-\infty}^\infty e^{-ax^2} e^{-2bx} dx = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{a}}$
- $\int_{-\infty}^\infty xe^{-a(x-b)^2} dx = b\sqrt{\frac{\pi}{a}}$
- $\int_{-\infty}^\infty xe^{-ax^2+bx} dx = \frac{\sqrt{\pi} b}{2a^{3/2}} e^{\frac{b^2}{4a}}$
- $\int_{-\infty}^\infty x^2 e^{-ax^2} dx = \frac{1}{2}\sqrt{\frac{\pi}{a^3}}$

TRIGONOMETRY FORMULA

- $\sin(A+B) = \sin A.\cos B + \cos A.\sin B$
- $\sin(A-B) = \sin A.\cos B - \cos A.\sin B$
- $\cos(A+B) = \cos A.\cos B - \sin A.\sin B$
- $\cos(A-B) = \cos A.\cos B + \sin A.\sin B$
- $\tan(A+B) = \dfrac{\tan A+\tan B}{1-\tan A\tan B}$
- $\tan(A-B) = \dfrac{\tan A-\tan B}{1+\tan A-\tan B}$
- $\cot(A+B) = \dfrac{\cot B.\cot A-1}{\cot B+\cot A}$
- $\cot(A-B) = \dfrac{\cot B.\cot A+1}{\cot B-\cot A}$
- $\sin C+\sin D = 2.\sin\left(\dfrac{C+D}{2}\right).\cos\left(\dfrac{C-D}{2}\right)$
- $\sin C-\sin D = 2.\cos\left(\dfrac{C+D}{2}\right).\sin\left(\dfrac{C-D}{2}\right)$
- $\cos C+\cos D = 2.\cos\left(\dfrac{C+D}{2}\right).\cos\left(\dfrac{C-D}{2}\right)$
- $\cos C-\cos D = -2\sin\left(\dfrac{C+D}{2}\right).\sin\left(\dfrac{C-D}{2}\right)$

  $= 2\sin\left(\dfrac{C+D}{2}\right).\sin\left(\dfrac{D-C}{2}\right)$
- $\sin 2x = 2\sin x.\cos x = \dfrac{2\tan x}{1+\tan^2 x}$
- $\cos 2x = \cos^2 x-\sin^2 x = 2\cos^2 x-1$

  $= 1-2\sin^2 x = \dfrac{1-\tan^2 x}{1+\tan^2 x}$
- $\tan 2x = \dfrac{2\tan x}{1-\tan^2 x}$
- $\sin 3x = 3\sin x-4\sin^3 x$
- $\cos 3x = 4\cos^3 x-3\cos x$
- $\tan 3x = \dfrac{3\tan x-\tan^3 x}{1-3\tan^2 x}$
- $a\sin x+b\cos x \longrightarrow$ Max. value $= +\sqrt{a^2+b^2}$

  Min. value $= -\sqrt{a^2+b^2}$
- Odd $F(x) \Rightarrow F(-x) = -F(x)$
- Even $F(x) \Rightarrow F(-x) = +F(x)$

$\cos(-x) = \cos x$	$\sec(-x) = \sec x$
$\sin(-x) = -\sin x$	$\operatorname{cosec}(-x) = -\operatorname{cosec} x$
$\tan(-x) = -\tan x$	$\cot(-x) = -\cot x$

- $2\sin A.\cos B = \sin(A+B)+\sin(A-B)$
- $2\cos A.\sin B = \sin(A+B)-\sin(A-B)$
- $2\cos A.\cos B = \cos(A+B)+\cos(A-B)$
- $-2\sin A.\sin B = \cos(A+B)-\cos(A-B)$
- $\sin(A+B).\sin(A-B) = \sin^2 A-\sin^2 B$
- $\cos(A+B).\cos(A-B) = \cos^2 A-\sin^2 B$

• $\sin^2\theta+\cos^2\theta = 1$	$1+\cos 2x = 2\cos^2 x$
• $\sec^2\theta-\tan^2\theta = 1$	$1-\cos 2x = 2\sin^2 x$
• $\operatorname{cosec}^2\theta-\cot^2\theta = 1$	$\tan x = \dfrac{1-\cos 2x}{\sin 2x}$

* General Solution of Trigonometry

- $\sin x = 0 \Rightarrow x = n\pi$
- $\cos x = 0 \Rightarrow x = (2n+1)\,\pi/2$
- $\tan x = 0 \Rightarrow x = n\pi$
- $\sin x = \sin y \Rightarrow x = n\pi+(-1)^n.y$
- $\cos x = \cos y \Rightarrow x = 2n\pi \pm y$
- $\tan x = \tan y \Rightarrow x = n\pi+y$
- $\sin^2 x = \sin^2 y$
- $\cos^2 x = \cos^2 y \quad \Rightarrow \quad x = n\pi \pm y$
- $\tan^2 x = \tan^2 y \quad \boxed{n \in \mathbb{Z}}$